Submitted by:

ADAN AADIL 2025_CYS_21

Assignment:

Lab No. 1

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

Department of Computer Engineering

University of Engineering and Technology, Lahore

Pacman Game Project Report

Project Title:
Pacman Game using Object-Oriented Programming (Java)

1. Introduction:
Pacman Game is a simple arcade-style game developed using Object-Oriented Programming (OOP) concepts in Java. In this game, the player controls Pacman, moves around the game board, collects food, and avoids being caught by the ghost. The project demonstrates the interaction of multiple classes and objects in a game environment.

2. Objectives:
- To implement Object-Oriented Programming concepts.
- To develop a simple console-based Pacman game.
- To demonstrate interaction between multiple classes and objects.
- To handle user input for Pacman movement.
- To implement game logic for winning and losing conditions.
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3. Technologies Used:
- Java
- Object-Oriented Programming (OOP)
- Scanner Class (for user input)
- VS Code / IntelliJ IDEA
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4. Classes Used:

4.1 PacmanGame Class
- Contains the main method.
- Initializes the game.
- Creates objects of GameBoard, Pacman, Ghost, and Food.
- Runs the game loop.
- Handles user input.

4.2 GameBoard Class
- Creates the game board.
- Displays the positions of Pacman, Ghost, and Food.
- Defines board dimensions.

4.3 Pacman Class
- Represents the player.
- Controls Pacman movement.
- Moves in four directions:
    - Up
    - Down
    - Left
    - Right

4.4 Ghost Class
- Represents the enemy.
- Automatically moves toward Pacman.

4.5 Food Class
- Represents food for Pacman.
- Determines the winning condition when collected.

5. Working Procedure:

1. The game starts.
2. The board is displayed.
3. The user enters movement commands (W, A, S, D).
4. Pacman moves according to input.
5. Ghost automatically moves toward Pacman.
6. If Pacman reaches food → Player Wins.
7. If Ghost catches Pacman → Game Over.

6. OOP Concepts Used:

Encapsulation

Data members (x and y coordinates) are private and accessed using getter methods.

Classes and Objects

Multiple objects interact in the game:

- Pacman object
- Ghost object
- Food object
- GameBoard object

Constructors

Constructors initialize object positions.

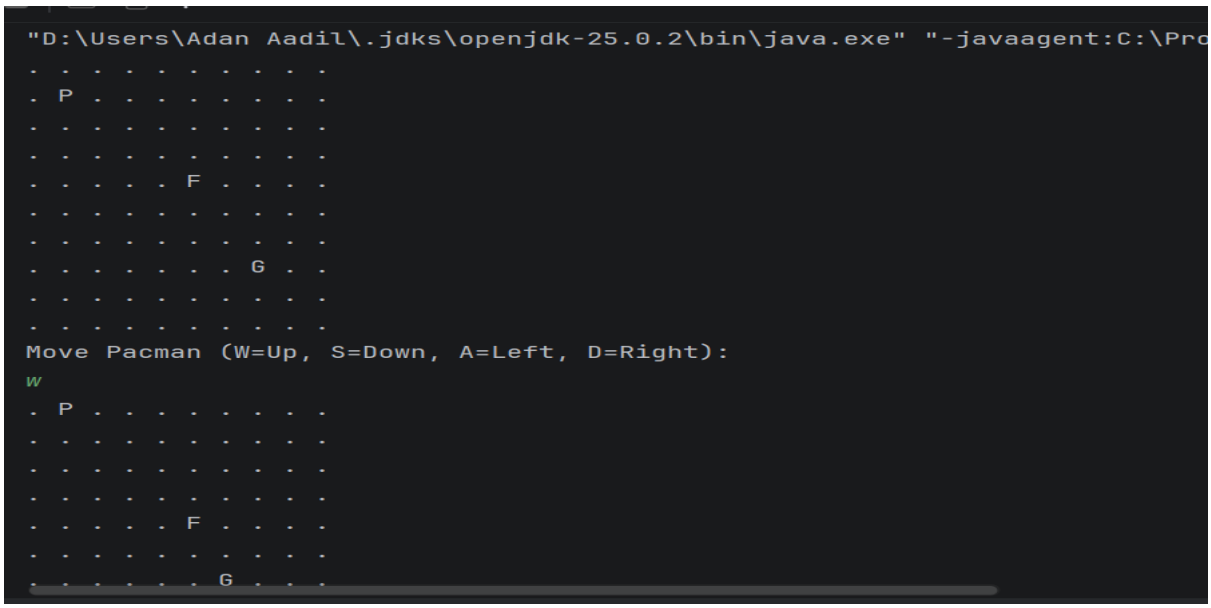
Method Interaction

Objects communicate through methods such as:

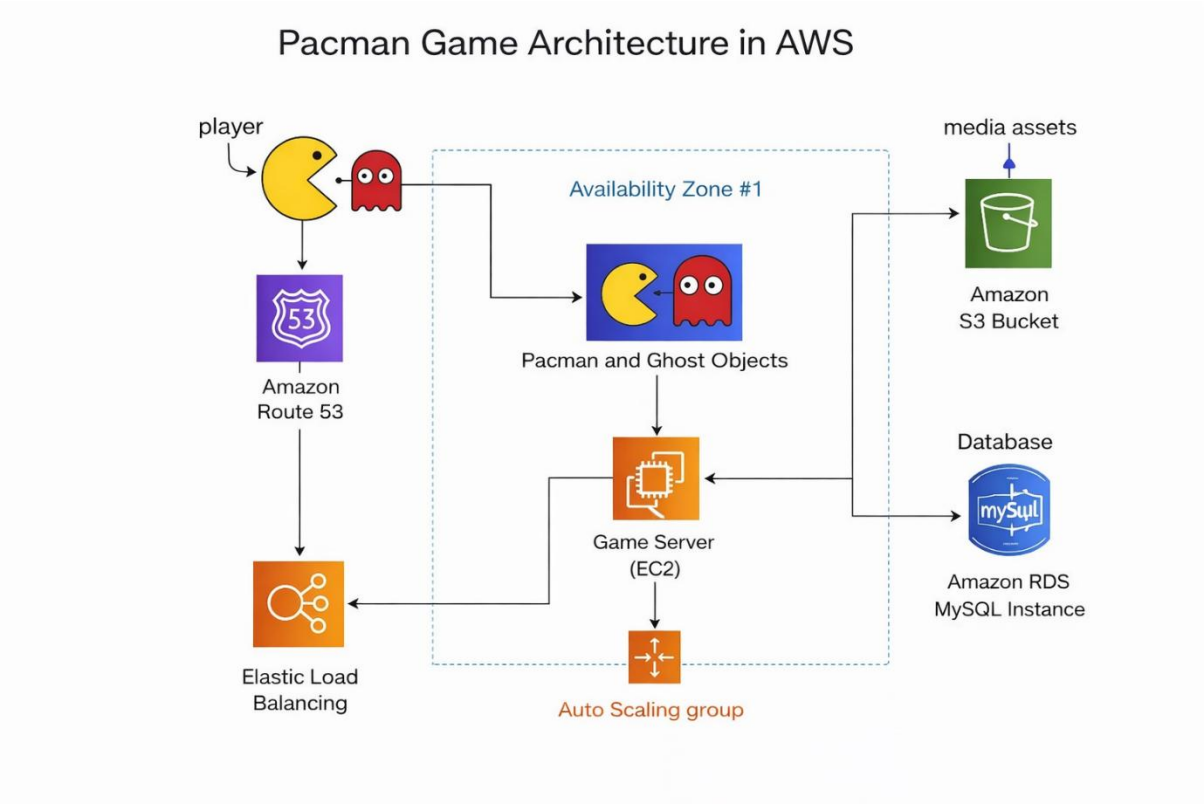
- move()
- moveTowardPacman()
- displayBoard()

8. Output

- Console-based game board is displayed.
- Pacman moves based on user input.
- Ghost chases Pacman.
- Win or Game Over message is displayed.



9. Conclusion

This project demonstrates the implementation of Object-Oriented Programming concepts through a simple Pacman game. It improves understanding of classes, objects, user interaction, and game logic in Java.

Architechtural diagram: